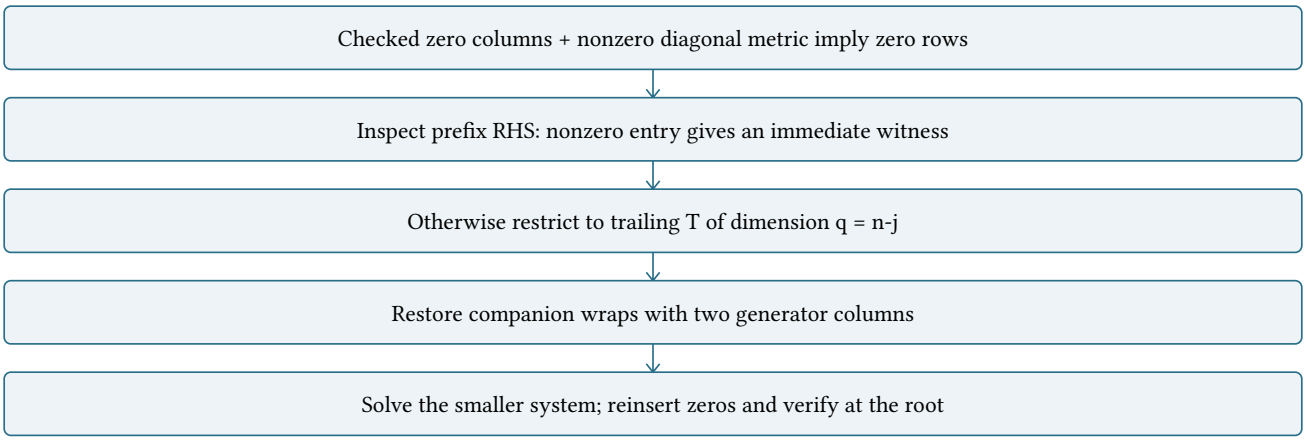
Zero-prefix coordinate deletion

2026-11-02 · Derived identities / static integration · Numerical trials: zero

Invariant. With nonzero diagonal D and $DA=A^{\mathsf{T}}D$, checked zero columns are also zero rows. A proper zero prefix of length j gives A=diag(0,T), where T has dimension q=n−j and inherits the trailing metric.



Displacement formula. Write S for the q-dimensional nilpotent shift, e for its first unit vector and f for its last. The trailing block gives $ST-TS=G_tK_t^{\mathsf{T}}$. Restoring distinct companion wraps yields

$G' = [G_t, a\,e, -b\,T\,e]$, $K' = [K_t, T^{\mathsf{T}} f, f]$.

Width grows from w to w+2 before recompression. The selector retains its first nonzero column to supply T e, including secondary-diagonal selection. One transpose source query supplies $T^{\mathsf{T}} f$. The identity also holds for q=1.

Prefix RHS	Derived branch
Has nonzero entry b_t	Witness e_t/b_t; zero new source queries
All entries zero	One transpose query; reduce n to n−j
No zero prefix	Unchanged paired pivot route

Exact reinsertion. Both solution and dual witness tails lift by zero padding. Prior Schur frames then unwind normally, and the original root checks remain. Early witnesses leave the trailing rank unknown: report a null full rank and only the established pivot-count lower bound.

Charged work. Reduction uses one original-width query, q field products and one subtraction outside it, plus generator copies and allocation. The next iteration repeats exact compression and selection on the smaller operator. Retaining the first column adds at most n field entries in the secondary-diagonal branch. ZeroFrame retains only dimensions, meter and a consumed flag; padding allocates j zeros and copies q tail entries. No net runtime or heap improvement is measured. Prefix frequency is unmeasured.

Checks and next. Three declared-substitution AST audits pass; six Python sources parse. A frame AST check excludes source/field-arithmetic calls. Numerical, corruption, heap and performance validation is unexecuted under the existing cost gate. Required controls include scalar tails, repeated deletions, alternating Schur/permutation frames, both RHS outcomes, independent displacement/lift checks and rank semantics. Next reuse the selected pivot columns on the reduced operator to avoid repeating search.

Sources. DERIVATION.md, deletion.py, selection_delete.py, driver_delete.py, integration.json. Standard cyclic-shift displacement conventions: Bostan et al.2017, SIAM J. Matrix Anal. Appl.38(3), equation (2).